

习题 1.3

1. 令 $\mathbf{Q}(i) = \{a + bi \mid a, b \in \mathbf{Q}\}$, 证明 $\mathbf{Q}(i)$ 是一个数域。

2. 令

$$F = \left\{ \frac{a_0 + a_1 e + \cdots + a_n e^n}{b_0 + b_1 e + \cdots + b_m e^m} \mid n, m \text{ 为任意非负整数}, a_i, b_j \in \mathbf{Z}, \right. \\ \left. 0 \leq i \leq n, 0 \leq j \leq m. \right\}$$

证明 F 是一个数域, 其中 e 是自然对数的底。

1. $1, 0 \in \mathbf{Q}(i)$ $\alpha = a + bi, \beta = c + di, \alpha \pm \beta = (a \pm c) + (b \pm d)i.$

$$\alpha \cdot \beta = (ac - bd) + (ad + bc)i.$$

若 $\beta \neq 0$, $\frac{\alpha}{\beta} = \frac{a + bi}{c + di} = \frac{(ac + bd)}{c^2 + d^2} + \frac{(bc - ad)}{c^2 + d^2}i.$

1. 解下述线性方程组:

$$\begin{cases} (1 + a_1)x_1 + x_2 + x_3 + \cdots + x_n = b_1, \\ x_1 + (1 + a_2)x_2 + x_3 + \cdots + x_n = b_2, \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ x_1 + x_2 + x_3 + \cdots + (1 + a_n)x_n = b_n, \end{cases}$$

其中 $a_i \neq 0, i = 1, 2, \dots, n$, 且 $\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} \neq -1$.

$$\begin{bmatrix} 1+a_1 & 1 & \cdots & 1 & b_1 \\ 1 & 1+a_2 & \cdots & 1 & b_2 \\ \vdots & \vdots & & \vdots & \vdots \\ 1 & 1 & \cdots & 1+a_n & b_n \end{bmatrix} \rightarrow \begin{bmatrix} a_1 & -a_2 & 0 & \cdots & 0 & b_1 - b_2 \\ 0 & a_2 & -a_3 & \cdots & 0 & b_2 - b_3 \\ 0 & 0 & a_3 & \cdots & 0 & b_3 - b_4 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -a_n & b_{n-1} - b_n \\ 1 & 1 & 1 & \cdots & 1+a_n & b_n \end{bmatrix} \quad \begin{cases} \text{令 } y = x_1 + x_2 + \cdots + x_n. \\ \therefore \begin{cases} y + a_1 x_1 = b_1 \\ y + a_2 x_2 = b_2 \\ \vdots \\ y + a_n x_n = b_n \end{cases} \rightarrow x_i = \frac{b_i - y}{a_i} \end{cases}$$

$$\rightarrow \begin{array}{cccccc} a_1 & 0 & 0 & \cdots & 0 & b_1 - b_n \\ 0 & a_2 & 0 & \cdots & 0 & b_2 - b_n \\ 0 & 0 & a_3 & \cdots & 0 & b_3 - b_n \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & a_{n-1} & -a_n \\ 1 & 1 & 1 & \cdots & 1 & 1+a_n \end{array} \quad \begin{array}{c} b_1 - b_n \\ b_2 - b_n \\ b_3 - b_n \\ \vdots \\ b_{n-1} - b_n \\ b_n \end{array}$$

$$\therefore y = \sum_{i=1}^n \frac{b_i}{a_i} - \left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} \right) \cdot y.$$

$$\text{令 } S = \frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n} + 1 \neq 0$$

$$\therefore Sy = \sum_{i=1}^n \frac{b_i}{a_i}, \quad y = \frac{1}{S} \cdot \sum_{i=1}^n \frac{b_i}{a_i}$$

$$\therefore x_i = \frac{b_i}{a_i} - \frac{1}{S \cdot a_i} \cdot \sum_{i=1}^n \frac{b_i}{a_i}$$

2. 解下述线性方程组:

$$\begin{cases} x_1 + 2x_2 + 3x_3 + \cdots + (n-1)x_{n-1} + nx_n = b_1, \\ nx_1 + x_2 + 2x_3 + \cdots + (n-2)x_{n-1} + (n-1)x_n = b_2, \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ 2x_1 + 3x_2 + 4x_3 + \cdots + nx_{n-1} + x_n = b_n. \end{cases}$$

全都加到第一行, $x_1 + x_2 + \cdots + x_n = \frac{2}{n(n+1)} \sum_{i=1}^n b_i$. 记作 s

①-③. 有 $(1-n)x_1 + x_2 + x_3 + \cdots + x_n = b_1 - b_2$

$$\therefore s - nx_1 = b_1 - b_2$$

类似的有:

$$s - nx_i = b_i - b_{i+1} \quad (i=1, 2, \cdots, n-1)$$

$$\therefore x_i = \frac{1}{n} \cdot (s + b_{i+1} - b_i)$$

$$x_n = s - \frac{1}{n} \cdot [(n-1)s + b_n - b_1]$$

$$= \frac{1}{n} \cdot s - \frac{1}{n} \cdot (b_n - b_1)$$

3. 解下述线性方程组:

$$\begin{cases} x_1 + x_2 + \cdots + x_n = 1, \\ x_2 + \cdots + x_n + x_{n+1} = 2, \\ \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \quad \cdots \\ x_{n+1} + x_{n+2} + \cdots + x_{2n} = n+1. \end{cases}$$

两两相减, 得

$$\begin{cases} x_{n+1} - x_1 = 1 \\ x_{n+2} - x_2 = 1 \\ \vdots \\ x_{2n} - x_n = 1 \\ x_1 + x_2 + \cdots + x_n = 1 \\ x_{n+1} + \cdots + x_{2n} = n+1 \end{cases}$$

$\rightarrow$

$$\begin{cases} x_{n+1} = x_1 + 1 \\ x_{n+2} = x_2 + 1 \\ \vdots \\ x_{2n-1} = x_{n-1} + 1 \\ x_{2n} = n+1 - x_1 - x_2 - \cdots - x_{n-1} - n+1 \\ = 2 - x_1 - x_2 - \cdots - x_{n-1} \end{cases}$$

共 $n-1$ 个自由变量

